

Prahlad Anand

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EDUCATION

Johns Hopkins University, Baltimore, MD	2024 – 2026
<i>Master of Science in Engineering, Computer Science</i>	GPA: 3.93/4.0
Vellore Institute of Technology, Vellore, India	2020 – 2024
<i>Bachelor of Technology, Computer Science and Engineering</i>	GPA: 8.86/10.0

EXPERIENCE

Image Analysis and Communications Lab, Johns Hopkins University
Graduate Research Assistant, Advised by Dr. Jerry Prince and Dr. Blake Dewey **Aug 2024 – Present**

- Developed a novel **vision-only isolated disentanglement framework** to learn a representation of image contrast.
- Used **self-supervised vision-language models** to **disentangle** anatomy and contrast representations from clinical brain MRIs and corresponding acquisition parameters. Improved on SOTA DICOM tag prediction accuracy by 15%.
- Adapted **Grad-CAM** to a CVAE-transformer network to analyze failure cases for brain MRI harmonization.
- Designed a **gradient-based adversarial attack** strategy using feature perturbation to improve test-time performance.

Artificial Intelligence and Robotics Lab, Indian Institute of Science, Bangalore, India
Research Assistant, Advised by Dr. Suresh Sundaram **Jan 2024 – Jun 2024**

- Trained a **semi-supervised RGB-to-infrared image translation** network, improving autonomous vehicle perception by 8.6% using synthetic data as augmentation over baseline unpaired image translation networks.
- Achieved a 4.2% performance improvement for semantic segmentation (DeepLabv3+) and object detection (YOLO, RCNN) models with **zero-shot** synthesis for simulation environments (**real-to-sim domain adaptation**).

Research Intern **May 2023 – Jul 2023**

- Enhanced the accuracy of **object detection and semantic segmentation** networks by 7% on RGB-IR datasets through multimodal fusion and **data augmentation using generative image translation** models (GANs, diffusion).
- Improved model accuracy by 10.2% over SOTA using supervised IR image synthesis and super-resolution.

Research Intern **May 2022 – Jul 2022**

- Experimented with complex-valued neural networks, loss functions, architectures, augmentation and hyperparameter tuning to boost multi-modal (EO-SAR) aerial image classification by 7.5% under severe class imbalance and noise.

PUBLICATIONS

- Anand, P.,** Carass, A., Mowry, E. M., Newsome, S. D., Prince, J. L., Dewey, B. E. (2026). *DICOM-CLIP: Zero-Shot Acquisition Parameter Retrieval from Unprocessed DICOM Files*. [ISBI](#). **(Oral Presentation)**
- Sikdar, A., Saadiyeen, Q., **Anand, P.,** Sundaram, S. (2024). *SSL-RGB2IR: Semi-supervised RGB-to-IR Image-to-Image Translation for Enhancing Vision Task Training in Semantic Segmentation and Object Detection*. [IROS](#).
- Anand, P.,** Saadiyeen, Q., Sikdar, A., N., Nalini, Sundaram, S. (2024). *Supervised Image Translation from Visible to Infrared Domain for Object Detection*. [arXiv](#).

SKILLS

Programming - C/C++, Python, PyTorch, TensorFlow, Java, JavaScript, MySQL, Bash, PHP, HTML/CSS
Technologies - OpenCV, numpy, pandas, HuggingFace, Unix/Linux, Git, Windows, Microsoft Office, E-mail
Deep/Machine Learning, Data Science, Medical Device, Computer Vision, NLP, RL, French (B2), German (A2)

PROJECTS

Neuroimage Synthesis Pipeline

- Skull-stripped brain MRIs using a U-Net. Performed **deformable registration** using the ANTs framework, and extended a CVAE-transformer harmonization network to perform modality fusion and cross-modal synthesis.

Product Recommendation System - [Link](#)

- Developed an inference pipeline to **web crawl Amazon and create real-time datasets using Selenium**. Suggests products based on **similarity in learned representations** of product images (CNN) and text metadata (BERT).

Video Sign Language Recognition

- Fine-tuned lightweight image classification networks** (ResNet50/ViT) to achieve +6.7% accuracy and 2.5× faster inference over VideoMAE (SOTA) on sign-language classification for videos using time-averaged predictions.

Dots: Geospatial Data Visualization for Businesses - [Link](#)

- Built Dots, a **geospatial business intelligence tool** integrating LLM-powered RAG with DuckDB/Postgres/Cesium to visualize and query data as map-based knowledge graphs. **Top-10 from 310 participants at HopHacks 2025**.

Token-Aware Hybrid Search for Reasoning in Large Language Models - [Link](#)

- Implemented **token-aware multi-step mathematical reasoning in LLMs** using a **Qwen-3B backbone**, a **Qwen-1.5B reward model**, and a **tree-search environment** with PPO fine-tuning and heuristic warm-start.